

Project Title: Myntra – E-Commerce database Management System Skill
DBMS – PAYMENT MANAGEMENT

1. Database Design

1.1 Payment Table

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK	References Orders
Payment_Method	VARCHAR(30)	-	UPI/Card/COD
Transaction_ID	VARCHAR(100)	-	Transaction ID
Payment_Status	VARCHAR(20)	-	Successful / Failed
Payment_Date	DATE	-	Date of payment
Amount	DECIMAL(10,2)	-	Payment amount

2. Table Relationships

Customer —→ **Orders = 1 : M**

Orders —→ **Payment = 1 : 1**

Orders —→ **Orders items = 1: M**

Product —→ **Orders items = 1: M**

Category —→ **Product = 1: M**

3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER(5) PRIMARY KEY,  
3     Order_ID NUMBER(5) NOT NULL,  
4     Payment_Method VARCHAR2(30) NOT NULL,  
5     Transaction_ID VARCHAR2(100) NOT NULL,  
6     Payment_Status VARCHAR2(20) NOT NULL,  
7     Payment_Date DATE NOT NULL,  
8     Amount NUMBER(10,2) NOT NULL,  
9     CONSTRAINT fk_payment_order  
10    FOREIGN KEY (Order_ID)  
11    REFERENCES Orders(Order_ID)  
12 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2  (Payment_ID, Order_ID, Payment_Method, Transaction_ID,  
3   Payment_Status, Payment_Date, Amount)  
4  VALUES  
5  (701, 501, 'UPI', 'UPI5012026',  
6   'Successful', TO_DATE('10-SEP-2026','DD-MON-YYYY'), 1300.00);
```

1 row created.

```
SQL> INSERT INTO Payment  
2  (Payment_ID, Order_ID, Payment_Method, Transaction_ID,  
3   Payment_Status, Payment_Date, Amount)  
4  VALUES  
5  (703, 503, 'UPI', 'UPI5032026',  
6   'Successful', TO_DATE('12-SEP-2026','DD-MON-YYYY'), 2799.20);
```

1 row created.

```
SQL> INSERT INTO Payment
  2  (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3  Payment_Status, Payment_Date, Amount)
  4  VALUES
  5  (702, 502, 'Card', 'CARD5022026',
  6  'Successful', TO_DATE('16-SEP-2026','DD-MON-YYYY'), 3398.30);

1 row created.

SQL> INSERT INTO Payment
  2  (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3  Payment_Status, Payment_Date, Amount)
  4  VALUES
  5  (704, 504, 'Card', 'CARD5042026',
  6  'Successful', TO_DATE('13-SEP-2026','DD-MON-YYYY'), 5699.05);

1 row created.

SQL> INSERT INTO Payment
  2  (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3  Payment_Status, Payment_Date, Amount)
  4  VALUES
  5  (705, 505, 'COD', 'COD5052026',
  6  'Successful', TO_DATE('14-SEP-2026','DD-MON-YYYY'), 1619.10);

1 row created.
```

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID
701	501	UPI	
UPI5012026			
Successful	10-SEP-26	1300	

703	503	UPI	
UPI5032026			
Successful	12-SEP-26	2799.2	

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID
702	502	Card	
CARD5022026			
Successful	16-SEP-26	3398.3	

704	504	Card	
CARD5042026			

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID
Successful	13-SEP-26	5699.05	

705	505	COD	
COD5052026			
Successful	14-SEP-26	1619.1	

5.2 Display Failed Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Failed';
```

no rows selected

```
SQL>
```

6. Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Successful'
  3 WHERE Payment_ID = 706;
```

1 row updated.

```
SQL>
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 706;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD
706	506	UPI
UPI5062026		
Successful	15-SEP-26	2462.24

7. Analyze Payment Methods

7.1 Calculate Total Transactions by Payment Method

```
SQL> SELECT
  2     Payment_Method,
  3     COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Method
  6 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
COD	1
Card	2
UPI	3

SQL>

7.2 Calculate Amount Collected by Payment Method

```
SQL> SELECT
  2     Payment_Method,
  3     SUM(Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Successful'
  6 GROUP BY Payment_Method
  7 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
COD	1619.1
Card	9097.35
UPI	6561.44

```
SQL>
```

8. Payment Transaction Report

```
SQL> SELECT
  2   p.Payment_ID,
  3   o.Order_ID,
  4   c.Customer_ID,
  5   c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  6   pr.Product_ID,
  7   pr.Product_Name,
  8   cat.Category_ID,
  9   cat.Category_Name,
 10   p.Payment_Method,
 11   p.Transaction_ID,
 12   p.Payment_Date,
 13   p.Amount,
 14   p.Payment_Status
 15 FROM Payment p
 16 JOIN Orders o
 17   ON p.Order_ID = o.Order_ID
 18 JOIN Customer c
 19   ON o.Customer_ID = c.Customer_ID
 20 JOIN Order_Items oi
 21   ON o.Order_ID = oi.Order_ID
 22 JOIN Product pr
 23   ON oi.Product_ID = pr.Product_ID
 24 JOIN Category cat
 25   ON pr.Category_ID = cat.Category_ID
 26 ORDER BY p.Payment_Date DESC;
```

```
PAYMENT_ID  ORDER_ID  CUSTOMER_ID
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
CATEGORY_ID  CATEGORY_NAME
```

```
-----
```

```
PAYMENT_METHOD
```

```
-----
```

```
TRANSACTION_ID
```

```
-----
```

```
PAYMENT_D      AMOUNT  PAYMENT_STATUS
```

```
-----
```

```
          702          502          102
```

```
-----
```

```
PAYMENT_ID  ORDER_ID  CUSTOMER_ID
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
CATEGORY_ID  CATEGORY_NAME
```

```
-----
```

PAYMENT_ID ORDER_ID CUSTOMER_ID

CUSTOMER_NAME

PRODUCT_ID

PRODUCT_NAME

CATEGORY_ID CATEGORY_NAME

PAYMENT_METHOD

TRANSACTION_ID

PAYMENT_D AMOUNT PAYMENT_STATUS

Priya Ravi

PAYMENT_ID ORDER_ID CUSTOMER_ID

CUSTOMER_NAME

PRODUCT_ID

PRODUCT_NAME

CATEGORY_ID CATEGORY_NAME

PAYMENT_METHOD

TRANSACTION_ID

PAYMENT_D AMOUNT PAYMENT_STATUS

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PAYMENT_ID ORDER_ID CUSTOMER_ID

CUSTOMER_NAME

PRODUCT_ID

PRODUCT_NAME

CATEGORY_ID CATEGORY_NAME

PAYMENT_METHOD

TRANSACTION_ID

PAYMENT_D AMOUNT PAYMENT_STATUS

Floral Dress

PAYMENT_ID ORDER_ID CUSTOMER_ID

